ANALYSIS REPORT

Wave 3.3 Curve-Aware Training Campaign Results

Wave 3.3 Campaign Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Wave 3.3` curve-aware training campaign. The campaign goal was to test whether adding curve-aware loss terms to the harmonic residual-offset architecture improves the scalar training surface before returning the candidates to official TE curve-first verification.

The campaign completed all planned entries

- `12` completed runs;
- `0` failed runs;
- `4` loss profiles;
- `3` required direction surfaces: `global`, `Fw`, and `Bw`.

The runner-level scalar first entry is `te_track2g_curve_aware_raw_centered_shape_fw`, but this is not a single deployment winner. The closeout keeps one best candidate for each required surface. Official TE curve verification remains a separate operator-approved step.

Campaign Artifacts

Artifact	Path
Campaign output	`output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08`
Leaderboard	`output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08/campaign_leaderboard.yaml`
Best-run pointer	`output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08/campaign_best_run.yaml`
Execution report	`output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08/campaign_execution_report.md`
Planning report	`doc/reports/campaign_plans/track_2/2026-06-08-18-01-40_track2g_curve_aware_training_campaign_plan_report.md`
Campaign package	`config/training/track2g_curve_aware_training/campaigns/2026-06-08_track2g_curve_aware_training_campaign`

Execution Summary

Field	Value
Campaign name	`track2g_curve_aware_training_campaign_2026_06_08`
Completed runs	12
Failed runs	0
Model type	`curve_aware_harmonic_residual_offset_probe`
Loss profiles	`pointwise_control`, `raw_centered_shape`, `raw_offset`, `full_curve_composite`
Tested surfaces	`global`, `Fw`, `Bw`
Runner-level scalar first entry	`te_track2g_curve_aware_raw_centered_shape_fw`
Program-level scalar winner changed	no

Directional Branch Results

Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
`global`	`full_curve_composite_global`	`full_curve_composite`	0.003345	0.003713	0.003616
`Fw`	`raw_centered_shape_fw`	`raw_centered_shape`	0.003181	0.003571	0.003251
`Bw`	`pointwise_control_bw`	`pointwise_control`	0.003430	0.003945	0.003749

These three rows are the branch-level closeout result. The `Fw` row is also the campaign scalar leader, but it does not replace the `global` or `Bw` branches.

Loss Profile Leaderboard

Rank	Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
1	`Fw`	`raw_centered_shape_fw`	`raw_centered_shape`	0.003181	0.003571	0.003251
2	`Fw`	`full_curve_composite_fw`	`full_curve_composite`	0.003260	0.003630	0.003320
3	`Fw`	`raw_offset_fw`	`raw_offset`	0.003279	0.003698	0.003328
4	`global`	`full_curve_composite_global`	`full_curve_composite`	0.003345	0.003713	0.003616
5	`global`	`raw_centered_shape_global`	`raw_centered_shape`	0.003350	0.003753	0.003636
6	`Fw`	`pointwise_control_fw`	`pointwise_control`	0.003371	0.003763	0.003291
7	`Bw`	`pointwise_control_bw`	`pointwise_control`	0.003430	0.003945	0.003749
8	`global`	`raw_offset_global`	`raw_offset`	0.003465	0.003829	0.003564
9	`Bw`	`raw_centered_shape_bw`	`raw_centered_shape`	0.003465	0.003998	0.003740
10	`Bw`	`raw_offset_bw`	`raw_offset`	0.003471	0.003992	0.003751
11	`Bw`	`full_curve_composite_bw`	`full_curve_composite`	0.003511	0.004113	0.003803
12	`global`	`pointwise_control_global`	`pointwise_control`	0.003587	0.004001	0.003607

Baseline Comparison

Surface	Wave 3.3 Best	Wave 3.3 MAE	Wave 3.2 Ref.	Ref. MAE	Delta
`global`	`full_curve_composite`	0.003345	`T2F-bis clean global`	0.003528	+5.20%
`Fw`	`raw_centered_shape`	0.003181	`T2F-bis harmonic Fw`	0.002862	-11.14%
`Bw`	`pointwise_control`	0.003430	`T2F-bis harmonic Bw`	0.003336	-2.83%

Scalar training metrics show a mixed result. The full composite loss improves the `global` Wave 3.3 scalar branch versus the previous Wave 3.2 global control. The `Fw` and `Bw` branches do not beat the strongest Wave 3.2 direction-specific harmonic candidates on scalar `test_mae` .

This does not yet decide curve-first promotion. Wave 3.3 was designed to test curve-aware behavior, so the next evidence must come from official TE Curve Verification Pipeline curve playback, overlays, offset diagnostics, and mean-centered shape checks.

Loss Interpretation

Loss Profile	Observed Signal	Interpretation
`pointwise_control`	Best `Bw` Wave 3.3 scalar result.	The curve-aware add-ons are not automatically beneficial on the backward branch.
`raw_centered_shape`	Best `Fw` Wave 3.3 scalar result and campaign scalar winner.	Centered-shape pressure is useful for forward scalar training, but not enough to beat Wave 3.2 harmonic `Fw` .
`raw_offset`	No branch winner.	Offset-only pressure did not dominate the first scalar campaign.
`full_curve_composite`	Best `global` Wave 3.3 scalar result.	Combining pointwise, centered-shape, offset, amplitude, and sparse harmonic terms is most promising for the global branch.

The main modeling result is not "Wave 3.3 wins". The useful result is narrower: the composite objective is promising for `global` , centered-shape is promising for `Fw` , and `Bw` still resists the first curve-aware loss profiles.

Registry Effects

The campaign runner refreshed family registries for all twelve Wave 3.3 families and updated the program registry. The program-level scalar best did not move: `te_periodic_gru_sequence_remote_Bw` remains the current scalar program winner with test `MAE = 0.002344` .

Registry Scope	BestRun	Test MAE
`Wave 3.3 global composite`	`full_curve_composite_global`	0.003345
`Wave 3.3 Fw centered-shape`	`raw_centered_shape_fw`	0.003181
`Wave 3.3 Bw pointwise`	`pointwise_control_bw`	0.003430

TE Curve Verification Pipeline Boundary

Official TE curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

The next TE Curve Verification Pipeline package should add all twelve Wave 3.3 candidates and report the accepted result separately for:

- `global` ;
- `Fw` ;
- `Bw` .

  The verification must compare raw curve error, centered-shape error, offset, amplitude, harmonic behavior, collage plots, and overlay plots against Wave 3.1, Wave 3.2, Wave 2.2, and the current accepted TE Curve Verification Pipeline baselines.

Closeout Decision

Wave 3.3 is complete from an execution standpoint: all planned candidates have successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry `full_curve_composite_global` forward as the strongest Wave 3.3 global scalar candidate;
- carry `raw_centered_shape_fw` forward as the strongest Wave 3.3 forward scalar candidate;
- carry `pointwise_control_bw` forward as the strongest Wave 3.3 backward scalar candidate;
- do not promote Wave 3.3 over Wave 3.2 or Wave 2.2 before official TE Curve Verification Pipeline verification.

Recommended Follow-Up

1. Accept this closeout and clear the active campaign state.
2. Prepare a separate operator-launched TE curve verification refresh for all

   twelve Wave 3.3 candidates.
3. Use the TE Curve Verification Pipeline result to decide whether Wave 3.3 should continue as a

   loss-only branch or whether the next campaign should move to the explicit multi-head shape/offset architecture.
4. Keep `Fw`, `Bw`, and `global` as parallel branch decisions.